

Curriculum Vitae: updated May 14, 2026

I. IDENTIFYING DATA

Name M. Brandon Westover, MD, PhD
Current Position Emily Fisher Landau Professor of Neurology, Harvard Medical School, Beth Israel Deaconess Medical Center (BIDMC)
Current Affiliation Beth Israel Deaconess Medical Center (BIDMC)

II. EDUCATION HISTORY

Colleges and Universities Attended

7/1999 BS in Physics, Brigham Young University
7/2006 PhD, Physics, Washington University in St. Louis
7/2006 MD, Washington University School of Medicine

Residency and Fellowship Training

2006 – 2007 Internship, Barnes Jewish Hospital, Washington University School of Medicine, St. Louis, MO
2007 – 2010 Residency, Partners Neurology Program (Massachusetts General Hospital and Brigham and Women's Hospital), Boston, MA
2010 – 2012 Fellowship, Epilepsy and Clinical Neurophysiology, Massachusetts General Hospital, Boston, MA

Board Certification

2011

III. EMPLOYMENT

Academic Appointments:

07/12-05/14 Instructor in Neurology, Harvard Medical School, Boston, MA
05/14-05/18 Assistant Professor in Neurology, Harvard Medical School, Boston, MA
05/18-10/22 Associate Professor
10/22 – present Professor of Neurology, Harvard Medical School, Boston, MA
11/22 – present Emily Fisher Landau Professor of Neurology, Harvard Medical School, Boston, MA

Other Appointments:

2022 to present Director, Critical Care EEG Monitoring Service, BIDMC

IV. HONORS AND AWARDS

2011 Best Paper Award Statistical Literacy (www.statlit.org) Statistical Education
2012 Best Paper Award Association for Computing Machinery: Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD) Basic Research
2014 Rappaport Fellowship Massachusetts General Hospital Nominated award
2018 #1 Most Disruptive AI Innovator Partners Healthcare Partners World Innovation Forum
2018 Innovation Discovery Grant Partners Innovation Office Artificial Intelligence Research / Development
2018 MGH-MIT Grand Challenge Winner ECOR Artificial Intelligence Research

2018	Best Neurology Clinical Research Abstract	MGH Neurology Department Clinical Research
2018	Best Paper Award CyberWorlds 2018, Singapore	Computer Science
2019	Disruptive Dozen AI Innovator	Massachusetts General Hospital Partners HealthcareWorld Innovation Forum
2019	Innovation Discovery Award (IDG)	Partners Healthcare World Innovation Forum
2019	Innovative Applications in Analytics Award (IAAA)	INFORMS Conference on Business Analytics & Operations ResearchFor achievements featuring novel combinations of descriptive, predictive and prescriptive analytics.
2019	Best Oral Abstract: Translational American Delirium Society (ADS)	ADS annual meeting
2019	Derek Denny-Brown Young Neurological Scholar Award	American Neurological Association
	ANA's Highest award. For outstanding scientific advances toward the prevention, diagnosis, treatment, and cure of neurological diseases	

V. BIBLIOGRAPHY

Peer-reviewed original research (370 total)

1. Eliasmith C, **Westover MB**, Anderson CH. "A general framework for neurobiological modeling: an application to the vestibular system." *Neurocomputing (Amst)*. 2002 Jun;44-46:1071-6. doi: 10.1016/s0925-2312(02)00418-6. PMID: PMC6788744.
2. **Westover MB**, Eliasmith C, Anderson CH. "Linearly decodable functions from neural population codes." *Neurocomputing (Amst)*. 2002 Jun;44-46:691-696. doi: 10.1016/s0925-2312(02)00459-9. Epub 2002 Mar 26. PMID: PMC7062372.
3. Fischer BJ, **Westover MB**. "The neural multiple access channel." *Neurocomputing (Amst)*. 2003 Jun;52-54:511-518. doi: 10.1016/s0925-2312(02)00762-2. Epub 2003 Jan 29. PMID: PMC7062373.
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5. **Westover MB**, O'Sullivan JA. "Achievable Rates for Pattern Recognition." *IEEE Trans Inf Theory*. 2008 Jan;54(1):299-320. doi: 10.1109/tit.2007.911296. Epub 2008 Jan 04. PMID: PMC7062371.
6. **Westover MB**. "Asymptotic Geometry of Multiple Hypothesis Testing." *IEEE Trans Inf Theory*. 2008 Jul;54(7):3327-3329. doi: 10.1109/TIT.2008.924656. Epub 2008 Jun 17. PMID: PMC6788803.
7. Woeltje KF, Butler AM, Goris AJ, Tutlam NT, Doherty JA, **Westover MB**, Ferris V, Bailey TC. "Automated surveillance for central line-associated bloodstream infection in intensive care units." *Infect Control Hosp Epidemiol*. 2008 Sep;29(9):842-6. doi: 10.1086/590261. PMID: PMC6788749.
8. Mueen A, Keogh E, Zhu Q, Cash S, **Westover B**. "Exact Discovery of Time Series Motifs." *Proc SIAM Int Conf Data Min*. 2009;2009:473-484. doi: 10.1137/1.9781611972795.41. PMID: PMC6814436.
9. Chu-Shore J, **Westover MB**, Bianchi MT. "Power law versus exponential state transition dynamics: application to sleep-wake architecture." *PLoS One*. 2010 Dec 02;5(12):e14204. doi: 10.1371/journal.pone.0014204. PMID: PMC2996311.
10. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**, Bigdely-Shamlo N. "A disk-aware algorithm for time series motif discovery." *Data Min Knowl Discov*. 2011 Jan;22(1-2):73-105. doi: 10.1007/s10618-010-0176-8. Epub 2010 Apr 18. PMID: PMC7062370.
11. Zhang WJ, **Westover MB**, Keary CJ. "Premortem diagnosis of sporadic Creutzfeldt-Jakob disease aided by positron-emission tomography imaging." *AJNR Am J Neuroradiol*. 2011 Jan;32(1):E18. doi: 10.3174/ajnr.A2292. Epub 2010 Nov 11. PMID: PMC6788750.
12. **Westover MB**, Bianchi MT, Eckman MH, Greenberg SM. "Statin use following intracerebral hemorrhage: a decision analysis." *Arch Neurol*. 2011 May;68(5):573-9. doi: 10.1001/archneurol.2010.356. Epub 2011 Jan 10. PMID: PMC3158138.
 - Featured article with accompanying editorial: *Arch Neurol*. 2011;68(5):565-566. "Statins After Intracerebral Hemorrhage: To Treat or Not to Treat".

- Covered in over 150 news feeds, including CBS News, ABC News, US News and World Report, and Medscape Medical News.
13. **Westover MB, Westover KD**, Bianchi MT. “Significance testing as perverse probabilistic reasoning.” BMC Med. 2011 Feb 28;9:20. doi: 10.1186/1741-7015-9-20. PMCID: PMC3058025.
 14. **Westover KD, Westover MB**, Winer EP, Richardson AL, Iglehart JD, Punglia RS. “Should a sentinel node biopsy be performed in patients with high-risk breast cancer?.” Int J Breast Cancer. 2011;2011:973245. doi: 10.4061/2011/973245. Epub 2011 Aug 10. PMCID: PMC3262582.
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 22. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover B**, Zhu Q, Zakaria J, Keogh E. “Searching and Mining Trillions of Time Series Subsequences under Dynamic Time Warping.” KDD. 2012 Aug;2012:262-270. doi: 10.1145/2339530.2339576. PMCID: PMC6816304.
 - Best Paper Award at Association for Computing Machinery’s annual SIGKDD (Special Interest Group on Knowledge Discovery and Data Mining) meeting, 2012.
 23. Eiseman NA, **Westover MB**, Ellenbogen JM, Bianchi MT. “The impact of body posture and sleep stages on sleep apnea severity in adults.” J Clin Sleep Med. 2012 Dec 15;8(6):655-66A. doi: 10.5664/jcsm.2258. PMCID: PMC3501662.
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 27. **Westover MB**, Cohen AB. “Reversible vasoconstriction syndrome with bilateral basal ganglia hemorrhages.” J Neuroimaging. 2013 Jan;23(1):122-5. doi: 10.1111/j.1552-6569.2011.00645.x. Epub 2011 Sep 07. PMCID: PMC6788746.

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Ching S, Liberman MY, Chemali JJ, **Westover MB**, Kenny JD, Solt K, Purdon PL, Brown EN. "Real-time closed-loop control in a rodent model of medically induced coma using burst suppression." *Anesthesiology*. 2013 Oct;119(4):848-60. doi: 10.1097/ALN.0b013e31829d4ab4. PMID: PMC3857134.
32. **Westover MB**, Bianchi MT, Yang C, Schneider JA, Greenberg SM. "Estimating cerebral microinfarct burden from autopsy samples." *Neurology*. 2013 Apr 09;80(15):1365-9. doi: 10.1212/WNL.0b013e31828c2f52. Epub 2013 Mar 13. PMID: PMC3662273.
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34. Recognized as work of special significance by Faculty of 1000: Baird A and Adamski M: F1000Prime Recommendation of [Westover MB et al., *Neurology* 2013, 80(15):1365-9]. In F1000Prime, 30 Apr 2013; DOI: 10.3410/f.718000104.793474863. F1000Prime.com/718000104#eval793474863.
 - Brandon **Westover M**, Shafi MM, Ching S, Chemali JJ, Purdon PL, Cash SS, Brown EN. "Real-time segmentation of burst suppression patterns in critical care EEG monitoring." *J Neurosci Methods*. 2013 Sep 30;219(1):131-41. doi: 10.1016/j.jneumeth.2013.07.003. Epub 2013 Jul 23. PMID: PMC3939433.
 - Ng MC, **Westover MB**, Cole AJ. "Treating seizures in Creutzfeldt-Jakob disease." *Epilepsy Behav Case Rep*. 2014;2:75-9. doi: 10.1016/j.ebcr.2014.01.004. Epub 2014 Feb 07. PMID: PMC4308028.
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42. Reviewed in *Epilepsy Currents*, official journal of the American Epilepsy Society: *Epilepsy*

43. Carlson C. "Can We Screen EEGs More Efficiently? Spectrographic Review of EEG Data." *Epilepsy Curr.* 2015;15(1):24-5. doi: 10.5698/1535-7597-15.1.24. PMID: PMC4320952.
 - Berkowitz AL, **Westover MB**, Bianchi MT, Chou SH. "Aspirin for acute stroke of unknown etiology in resource-limited settings: a decision analysis." *Neurology.* 2014 Aug 26;83(9):787-93. doi: 10.1212/WNL.0000000000000730. Epub 2014 Jul 23. PMID: PMC4155044.
 - Gaspard N, Hirsch LJ, LaRoche SM, Hahn CD, **Westover MB**, Critical Care EEG Monitoring Research Consortium. "Interrater agreement for Critical Care EEG Terminology." *Epilepsia.* 2014 Sep;55(9):1366-73. doi: 10.1111/epi.12653. Epub 2014 Jun 02. PMID: PMC4879939.
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 - Bianchi MT, Lipoma T, Darling C, Alameddine Y, **Westover MB**. "Automated sleep apnea quantification based on respiratory movement." *Int J Med Sci.* 2014;11(8):796-802. doi: 10.7150/ijms.9303. Epub 2014 May 30. PMID: PMC4057486.
45. Jing Jin, Dauwels J, Cash S, **Westover MB**. "SpikeGUI: software for rapid interictal discharge annotation via template matching and online machine learning." *Annu Int Conf IEEE Eng Med Biol Soc.* 2014;2014:4435-8. doi: 10.1109/EMBC.2014.6944608. PMID: PMC4416962.
46. Akeju O, **Westover MB**, Pavone KJ, Sampson AL, Hartnack KE, Brown EN, Purdon PL. "Effects of sevoflurane and propofol on frontal electroencephalogram power and coherence." *Anesthesiology.* 2014 Nov;121(5):990-8. doi: 10.1097/ALN.0000000000000436. PMID: PMC4206606.
47. Akama-Garren EH, Bianchi MT, Leveroni C, Cole AJ, Cash SS, **Westover MB**. "Weighing the value of memory loss in the surgical evaluation of left temporal lobe epilepsy: a decision analysis." *Epilepsia.* 2014 Nov;55(11):1844-53. doi: 10.1111/epi.12790. Epub 2014 Sep 19. PMID: PMC4877127.
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51. Featured article with accompanying editorial: "Early epileptiform discharges and the yield of prolonged EEG monitoring." Dangayach N, Claassen J., *Clinical Neurophysiology* (2014), doi: <http://dx.doi.org/10.1016/j.clinph.2014.07.002>.
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54. Kenny JD, **Westover MB**, Ching S, Brown EN, Solt K. "Propofol and sevoflurane induce distinct burst suppression patterns in rats." *Front Syst Neurosci.* 2014;8:237. doi: 10.3389/fnsys.2014.00237. Epub 2014 Dec 18. PMID: PMC4270179.
55. Ng MC, Gaspard N, Cole AJ, Hoch DB, Cash SS, Bianchi M, O'Rourke DA, Rosenthal ES, Chu CJ, **Westover MB**. "The standardization debate: A conflation trap in critical care electroencephalography."

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 60. Alvarez V, Drislane FW, **Westover MB**, Dworetzky BA, Lee JW. "Characteristics and role in outcome prediction of continuous EEG after status epilepticus: A prospective observational cohort." *Epilepsia*. 2015 Jun;56(6):933-41. doi: 10.1111/epi.12996. Epub 2015 May 07. PMCID: PMC4878827.
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Peer-reviewed publications (accepted, in press)

1. Nascimento FA, Katyal R, Kass NR, Yuan D, Sirven JI, Westover MB, Beniczky S. "Electroencephalography in clinical practice: results from an AAN Practice Current survey." *Journal of Clinical Neurophysiology*.**
2. Sartipi S, Godfrey DS, Tauţan AM, Fernandes MP, Ghanta M, Gupta A, Nearing B, Kim J, Struck AF, Loddenkemper T, Peters J, Lee JW, Westover MB, Zafar SF. Automated Extraction of Seizures and Ictal-Interictal Continuum Patterns from EEG Reports to Enable Large-Scale Neurophysiology and Neurocritical Care Research. Available at SSRN. doi: 10.2139/ssrn.5489575. SSRN.**

Peer-reviewed publications (other - 8 total)

1. Westover MB. "Review of Peter Dayan & L. F. Abbott's Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems." *Philosophical Psychology*. 2002. 15: 563-567.
2. Bianchi MT and Westover MB. (2009) "Comment on: Analysis of SPARCL trial data for reduction of future stroke risk in older stroke patients with recent stroke or TIA." *Neurology* 73(10):817-818.
3. Westover MB, Gaspard N. "Not a simple plumbing problem: Updating our understanding of delayed cerebral ischemia in aneurysmal subarachnoid hemorrhage." *J Clin Neurophysiol*. 2016 Jun;33(3):171-3. PMID: PMC4894327. Invited review as Guest Editor for a special issue of JCN on the neurophysiology of subarachnoid hemorrhage.
4. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M, Schiefecker A, Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C, Piilgaard H, Rosenthal ES, Westover MB, Maslarova A, Santos E, Hertle D, Sanchez-Porrás R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Bohner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A, Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Guerresir, E, Vatter H, Chung L, Brennan K, Lieutaud T, Marinesco S, Maas A, Sahuquillo J, Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading

depolarizations in neurointensive care: review and recommendations of the COSBID research group.” *Journal of Cerebral Blood Flow and Metabolism*. 2016 Jun 17. pii: 0271678X16654496. [Epub ahead of print] PMID: 27317657.

5. Westover MB, Halford JJ, Bianchi MT. What it should mean for an algorithm to pass a statistical Turing test for detection of epileptiform discharges. *Clin Neurophysiol*. 2017 Jul;128(7):1406-1407. doi: 10.1016/j.clinph.2017.02.026. Epub 2017 Mar 16. PMID: 28495216; PMCID: PMC9359665.
6. Bianchi MT, Thomas RJ, Westover MB. An open request to epidemiologists: please stop querying self-reported sleep duration. *Sleep Med*. 2017 Jul;35:92-93. doi: 10.1016/j.sleep.2017.02.001. Epub 2017 Feb 17. PMCID: PMC9371612.
7. Singla S, Garcia GE, Rovenolt GE, Soto AL, Gilmore EJ, Hirsch LJ, Blumenfeld H, Sheth KN, Omay SB, Struck AF, Westover MB, Kim JA. “Detecting Seizures and Epileptiform Abnormalities in Acute Brain Injury.” *Curr Neurol Neurosci Rep*. 2020 Jul 27;20(9):42. doi: 10.1007/s11910-020-01060-4. Review. PMC9353690.
8. Nasiri S, Ganglberger W, Sun H, Thomas RJ, Westover MB. Exploiting Labels from Multiple Experts in Automated Sleep Scoring. *Sleep*. 2023 Feb 16:zsad034. doi: 10.1093/sleep/zsad034. Epub ahead of print. PMID: 36795078.

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Non-Peer-reviewed Articles

1. Westover MB, O’Sullivan JA. “Towards an information theoretic framework for object recognition.” IEEE 2004 International Symposium on Information Theory, Chicago, Illinois, June 27-July 2, 2004.
2. Westover MB, O’Sullivan JA. “Achievable rates for pattern recognition: Binary and Gaussian cases.” IEEE 2005 International Symposium on Information Theory, Adelaide, Australia, Sept 4-9, 2005.
3. O’Sullivan JA, Singla N, and Westover MB. "Successive Refinement for Pattern Recognition," IEEE Information Theory Workshop, Punta del Este, Uruguay, Feb 2006.
4. Liddle S, Zhang R, Grover L, Khitrov M, Brown J, Cobb JP, Goldman J, Chou J, Westover MB, Yagoda D, Reisner A, “Safety Evaluation of a Medical Device Data System”. International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'12), San Diego, California, USA on August 28 - September 1, 2012.
5. Bianchi MT, Thomas RJ, Westover MB. “Response.” *Sleep Med*. 2017 Aug 3. pii: S1389-9457(17)30316-7. doi: 10.1016/j.sleep.2017.07.017. [Epub ahead of print] PMID: 2884338
6. Amorim E, Zheng WL, Ghassemi MM, Aghaeiaval M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, Westover MB. “The International Cardiac Arrest Research (I-CARE) Consortium Electroencephalography Database.” *medRxiv*. 2023 Aug 28:2023.08.28.23294672. doi: 10.1101/2023.08.28.23294672. Preprint. PMID: 37693458.
7. Sun H, Ganglberger W, Westover MB, Thomas RJ. "Sleep as a Window into Brain Health: Clinical Implications from the Sleeping Brain Wave-Structure Associations." *SLEEP*. 2025 May. (Editorial).**

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Book Chapters

1. Westover MB, Thomas TT, Greer DM, “Delirium” in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 86-89.

2. Westover MB, Choi E, Awad KM, Greer DM, Editors. Pocket Neurology. Lippincott Williams & Wilkins, 2010.
3. Westover MB, Peters JM, Bromfield EB, “Seizures and Other Spells” in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 57-79.
4. Westover MB, Xia, Z, Cho TA, “Meningitis, Encephalitis, & Brain Abscesses” in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 136-149.
5. Thomas TT, Westover MB, “Rapidly Progressive Dementia” in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 89-96.
6. Kummer TT, Westover MB, “Central Nervous System Vasculitis” in Pocket Neurology, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 175-177.
7. Westover MB, Awad K, "Ischemic Stroke, Muscular Dystrophy and Friedreich's Ataxia" in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
8. Westover MB, Edlow BL, Greer DM, “Coma After Cardiac Arrest: Management and Neurological Prognostication” in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
9. Westover MB, Shafi MM, “Seizures” in Pocket ICU, Lippincott Williams & Wilkins, 2013.

VI. GRANT FUNDING

Current:

09/2021 – 05/2026	Funder: NIH-NINDS (R01 AG073410) Title: Comparative Safety of Seizure Prophylaxis within the Medicare Program Role: Co-Investigator (PI: Lidia Maria Moura)
05/2021 – 04/2026	Funder: NIH-NINDS (R01 NS120947) Title: Establishing a Brain Health Index from the Sleep Electroencephalogram Role: PI
09/2022 – 08/2027	Funder: NIH-NHLBI (R01 HL161253) Title: Data-Driven Sleep Biomarkers of Brain Health, Heart Health, and Mortality Role: PI
08/2022 – 04/2027	Funder: NIH-NIA (R01 AG073598) Title: Low Neurophysiologic Resistance to Anesthetics as a Marker of Preclinical/Prodromal Alzheimer's Disease and Neurovascular Pathology, Delirium Risk and Inattention Role: Co-Investigator (PI: Miles Berger, via Duke subcontract)
01/2023 – 12/2027	Funder: NIH-NINDS (R01 NS126282) Title: Prospective Validation of Neurophysiologic Outcome Prediction in Acute Brain Injury Role: PI
09/2023 – 05/2028	Funder: NIH-NINDS (R01 NS130119) Title: Late-Onset Unexplained Epilepsy as a Risk Factor for Dementia Role: Co-PI (with Alice D. Lam)
12/2023 – 11/2028	Funder: NIH-NINDS (R01 NS131347) Title: Comparative Effectiveness of EEG-Guided Anti-Seizure Treatment in Acute Brain Injury Role: Co-Investigator (PI: Sahar F. Zafar)

11/2023 – 10/2028	Funder: NIH-NIA (R01 AG078261) Title: Delirium Dynamics: Understanding Causes and Effects Role: Co-Investigator (PI: Eyal Kimchi, via Northwestern subcontract)
09/2023 – 09/2025	Funder: NIH-NIA (R21 AG085495) Title: Development of Novel Polysomnography-Based Digital Biomarkers to Predict Alzheimer's Disease and Parkinson's Disease Role: Co-Investigator (PI: Yue Leng, via UCSF subcontract)
02/2025 – 02/2026	Funder: Takeda (PO8001791544) Title: Analysis of Risk of Narcolepsy through Clinical Notes Role: PI
Submitted:	
12/2025 – 11/2030	Funder: NIH (Pending) Title: Leveraging Clinical Data and Automation to Predict Epilepsy After TBI Role: Co-Investigator (PI: Jennifer Kim)
12/2025 – 11/2030	Funder: NIH (Pending) Title: De-Biasing Effect Assessment of Seizure Effects on Neurologic Outcomes (DEBIAS) Role: Co-Investigator (PI: Adrian Dalca)
12/2025 – 11/2030	Funder: NIH-NINDS (Pending) Title: Automating Critical Care Electroencephalography Using AI Foundation Models Role: PI
12/2027 – 11/2030	Funder: NIH (Pending) Title: *Tana Mu Na?* Feasibility of Mobile EEG and Automated Intelligence to Diagnose Epilepsy in Children in Guinea-Bissau Role: Co-Investigator (PI: Farrah Mateen)
07/2025 – 06/2030	Funder: NIH-NINDS (1UG3NS143749-01) Title: Continuous Monitoring of Pain Using Facial Expressions in the ICU Environment (COMPANION) Role: Co-Investigator (PI: Hao Deng)
Completed:	
2012 – 2014	Funder: American Brain Foundation Clinical Research Training Fellowship Title: Early Detection of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage Using Context-Sensitive Computational EEG Role: PI
2014 – 2019	Funder: NIH-NINDS (K23 NS090900) Title: Quantitative Neurophysiological Monitoring and Control of Sedation-Analgesia in the ICU Environment Role: PI
2017 – 2019	Funder: SAGE Therapeutics

Title: Automated Detection of Seizure-Spectrum Patterns and Quantitative
Neurological Outcome Prediction
Role: PI

2018 – 2019 Funder: Partners Healthcare Innovation
Title: Artificial Intelligence-Based Seizure Detection and Classification
Role: PI

2018 – 2019 Funder: MGH-MIT Grand Challenge
Title: Big Data and Deep Learning for the Interictal-Ictal-Injury Continuum
Role: PI

2018 – 2021 Funder: Glenn Foundation for Medical Research
Title: Electroencephalogram-Based Brain Age and Its Relationship with
Cognitive Function and Sleep Quality
Role: PI

2018 – 2023 Funder: NIH-NINDS (R01 NS102190)
Title: Investigation of Sleep in the Intensive Care Unit
Role: PI

2019 – 2024 Funder: NIH-NINDS (R01 NS107291)
Title: Deep Learning for the Ictal-Interictal-Injury Continuum
Role: PI

2018 – 2023 Funder: NIH-NINDS (R01 NS102574)
Title: Multimodal Network Connectivity Architecture (MOCA) of the Brain and
Its Role in Recovery of Consciousness in Comatose Cardiac Arrest Patients
Role: Co-Investigator (PI: Ona Wu)

2019 – 2024 Funder: NIH-NIA (R01 AG060987)
Title: Treating Hyperexcitability in Alzheimer's Disease with Levetiracetam to
Improve Brain Function and Cognition
Role: Co-Investigator (PI: Mouhsin Shafi)

2019 – 2024 Funder: NIH-NIA (RF1 AG064312)
Title: Integrated Motor Activity Biomarker for Risk of Alzheimer's Dementia
Role: Co-Investigator (PI: Kun Hu)

VII. PROFESSIONAL ACTIVITIES

Editorial Service

2010 Editor, Pocket Neurology, Lippincott Williams & Wilkins,
2016 Editor, Pocket Neurology, 2nd Edition; Wolters Kluwer,
2019-2024 Editorial Board Member, Journal of Clinical Neurophysiology,

Ad Hoc Reviewer

2010- present
2010- present
2006- present
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2010- present

Service as Grant Reviewer

2018-2020 Ad Hoc Member, Acute Neural Injury & Epilepsy (ANIE), NIH
2021- 2025 Permanent Member, Acute Neural Injury & Epilepsy (ANIE), NIH

University Administrative Service

2011-2017
2011-2022
2011-2022
2010-2016
2013-2017
2014-2022
2015-2018 Member, PhD Thesis Committee (Candidate: Mohammad Ghassemi), MIT
2015-2019 Member, PhD Thesis Committee (Candidate: Jing Zhi An), MIT
2017-2019
2018-2020

Service to Professional Organizations

2011-2014 TRENdS (Treatment of Recurrent Nonconvulsive SeizureS) Clinical Trial Investigators Committee, Duke University / Duke Clinical Research Institute (DCRI)
2015-2017 Member at Large Critical Care EEG Monitoring Research Consortium (CCEMRC)
2016 Co-Chair, 7th International Conference on Extreme Learning Machines (ELM2016). Singapore, December 13-15, 2016.
2017-2019 Vice-Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2017 Co-Chair, The 8th International Conference on Extreme Learning Machines (ELM2017). Yantai, China, October 4-7, 2017.
2019-2021 Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2025-2029 Chair, International League Against Epilepsy (ILAE) – Commission on Big Data and AI
2025- Committee Member, NINDS Common Data Elements (CDE) in Epilepsy Research Oversight Committee.

Community Service

2012 Christ Lutheran Nursery School, Belmont, MA
2014 Boy Scout Troop 66, Belmont, MA. Belmont, MA

Educational Speaker
Educational Speaker

VIII. INVITED PRESENTATIONS (SINCE 2020)

Grand Rounds:

Local Invited Presentations:

National and Regional Meetings:

International Meetings:

IX. TEACHING

Teaching during Stanford appointment:

None.

Teaching prior to Stanford:

Courses

Tutorials and Lectures

Clinical Supervisory and Training

Research Supervisory and Training Responsibilities

2001–2002	Instructor for Computational Neuroscience, Washington University School of Medicine, Department of Anatomy and Neurobiology.
2003	Instructor for Detection and Estimation Theory, Washington University, Department of Electrical and Systems Engineering.
2004	Instructor for Information Theory, Washington University, Department of Electrical and Systems Engineering.
2008–2009	Instructor for Neuroanatomy Laboratory, Harvard Medical School.
2012–2025	Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.
2013–2017	Course Director and Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.
10/2009	Granger Causality Analysis Techniques for EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
5/2010	Synchronization Likelihood for Functional Connectivity Analysis of EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
7/2008	Probability in Medical Reasoning. Neurology Residents, Brigham and Women’s Hospital, Boston, MA.
9/2009	Invited lecturer, Bayes’ Rule for Physicians. Partners Neurology, MGH/BWH, Boston, MA.
6/2010	EEG Alphabet Soup: GCSE, NCSE, PLEDs, BIPLEDs, GPEDs, & DEADs. Partners Neurology, MGH/BWH, Boston, MA.
8/2011	Status Epilepticus: Diagnosis and Management. Partners Neurology, MGH/BWH, Boston, MA.
9/2012	Introduction to ICU EEG: Seizures, Spikes, PEDs, Ischemia, and Coma. Partners Neurology, MGH/BWH, Boston, MA.

7/2018	Quantitative Electroencephalography. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
7/2019	Recognizing Seizures and Ischemia in Critical Care EEG Monitoring. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
6/2020	Emergency Electroencephalography. Emergency Medicine Residents, Massachusetts General Hospital, Boston, MA.
7/2020	Introduction to Spectrograms in ICU EEG. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
2012–	Supervisor, MGH Adult Epilepsy Clinic (½-day session per week).
2012–	Attending Physician, MGH Inpatient Neurology Service (2-week block per year), supervising neurology residents and Harvard medical students.
2012–	Attending Physician, MGH General Neurology Consult Service (2-week block per year), supervising neurology residents.
2012–	Director, MGH Critical Care EEG Monitoring Service (≈10 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Attending, MGH Epilepsy Monitoring Unit (≈6 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Faculty, Outpatient EEG Reading Service (≈2 days per month), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Supervisor of postdoctoral research fellows (average 5–10 fellows per year), Massachusetts General Hospital.
2012–	Supervisor of graduate students (average 2–4 per year), Massachusetts General Hospital and MIT.

X. TRAINEES

Postdoctoral Fellows

Sahar Zafar, MD (K23)	2015–present
Jennifer Kim, MD, PhD (K23)	2014–2020
Edilberto Amorim, MD (K23)	2014–2019
Marta Priscila Bento Fernandes, PhD	2019–present
Rajesh Amerineni, PhD	2020–2022
Wendong Ge, PhD	2019–2022
Weilong Zheng, PhD	2019–2022
Yalda Amidi, PhD	2022–2024
Samaneh Nasiri, PhD	2021–2023
Wolfgang Ganglberger, PhD	2019–present
Shadi Sartipi, PhD	2024–present
Alexandra Tautan, PhD	2024–present
Ayush Tripathi, PhD	2024–present
Samuel Waters, PhD	2024–present
ChenXi Sun, PhD	2024–present
Charlotte Maschke, PhD	2024–present
Bruce Nearing, PhD	2024–present
Ayan Mitra, PhD	2024–present
Sungtae An, MS / PhD	2016–2018
Haoqi Sun, MD	2016–present
Jing Jin, PhD	2013–present

Graduate Students

Jingzhi An, PhD	2014–2020
Mohammad Ghassemi, PhD	2014–2019
Jefferson T. Oliva, PhD	2016–2017
Qiang Li, PhD	2018–2020
Arnav Gupta	2024–present
Thijs Nassi, MSc	2021–2025

Medical Students / Residents / Fellows

Elliott Akama-Garren	2014–2015
Jennifer Kim, MD, PhD	2014–2020
Eyal Kimchi, MD, PhD (K23)	2018–present
Daniel Goldenholz, MD, PhD (K23)	2022–present
Peter Hadar, MD, PhD	2022–present
Wan Yee Kong, MD, MSc	2022–2025
James Lucarelli, MD, PhD	2022–present
Callison Alcott, MD, PhD	2023–present
Sarah Wahlster, MD	2011–2012
Craig Williamson, MD	2011–2012
Lidia Moura, MD	2013–2014
Marcus Ng, MD	2013–2014
Patrick Chen, MD	2013–2014
Deirdre O’Rourke, MD	2013–2014
Sandipan Pati, MD	2013–2014
Ammar Kheder, MD	2013–2014
Aaron Berkowitz, MD, PhD	2013–2014
Luis Paixao, MD	2017–2019
Aline Herlopian, MD	2017–present
Nitish Harid, MD	2020–2021

Undergraduate Students

Nathaniel Eiseman	2010–2011
Wade Whitt	2017–2019
Jacob Hogan	2018–2020
Alek Westover	2018–2023
Danny Jones	2021–2022
Jaden Barfuss	2021–2022
Jayme Banks	2021–2023
Leia Jekel	2023–2024
Rebecca Milde	2023–2024
Moritz Alkofer	2023–2024
Elijah Davis	2023–2024
Dan Sumsion	2023–present
Lydia Petersen	2023–present
Kiwi (Yiwen) Xiong	2023–present
Luke Jackson	2024–present
Carlos Nuñez-Sosa	2024–present
Amber Peng	2024–present
Mitchell McCauley	2024–present
Patrick Wynn	2024–present
Elijah Keldsen	2024–present

High School Students

Sophia Bechek	2016–2017
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